TEXORA · LECTURE SHEET

Material Management and Sewing Room Management

Two management systems that meet at the bundle ticket: materials — buying, storing, issuing and reconciling fabric and trims against consumption standards — and the sewing room — line set-up, WIP discipline and hourly control of the factory's largest cost centre.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 6 · LEVEL 4 TERM 2

1. Material management: scope and stakes

Materials are 60–70% of a garment's cost (Topic 1), bought against orders months before shipment, in a market of shade lots, widths and lead times. Material management covers the full chain: **requirement calculation** (from markers and consumption standards), **procurement support** (quantities, dates, specs to merchandising/commercial), **receiving and inspection** (Topic 5), **storage** (identified, shade-lotted, FIFO, protected), **issue** (against cut plans only), and **reconciliation** (bought vs issued vs consumed vs leftover, per order). Its two failure modes are symmetric and both expensive: **shortage** (a line stops, an order ships short) and **surplus** (dead stock — fabric bought with the order's money that no order will ever consume).

2. Requirement and consumption control

WORKED EXAMPLES — VERIFY EVERY NUMBER (THE TOPIC 5 ORDER CONTINUED)

(a) Fabric. Order 12,000 garments, budgeted consumption 1.236 m/garment (Topic 5) → requirement = 12,000 × 1.236 = **14,832 m**. The store issues against the cut plan in lay quantities; after cutting, reconciliation compares metres issued with garments cut × 1.236 — variance beyond the allowance is investigated while the order is still in the building.

(b) Sewing thread. Total seam length per garment (from the operation bulletin) = 6.20 m, sewn in stitch type 301 with thread consumption ratio 2.5 (metres of thread per metre of seam, both needle and bobbin threads combined) → thread per garment = 6.20 × 2.5 = **15.5 m**. Add 10% for wastage, thread breaks and cone ends → 15.5 × 1.10 = **17.05 m**. For 12,000 garments: 12,000 × 17.05 = 204,600 m → cones of 5,000 m: 204,600 ÷ 5,000 = 40.9 → **41 cones** of the main shade.

> Thread looks trivial until a line stops for want of one cone of the right shade — consumption standards exist for every trim, not just fabric.

CONTROL	HOW IT WORKS
ABC classification	Store attention follows value: A items (fabric, major trims) counted and reconciled per order; C items (buttons by gross, polybags) controlled by bulk issue and periodic count.
Issue discipline	Nothing leaves the store without a requisition tied to a cut plan or trim card; partial rolls are measured out and back.
Leftover management	Order leftovers are measured, labelled (style, shade, width, length) and listed; recuts, replacements and sample work draw on them before new purchase; aged leftovers are reported, not forgotten.
Supplier data loop	Inspection results (short length, narrow width, shade variation) are recorded per supplier and roll — the evidence base for claims and for future sourcing.

3. Sewing room management

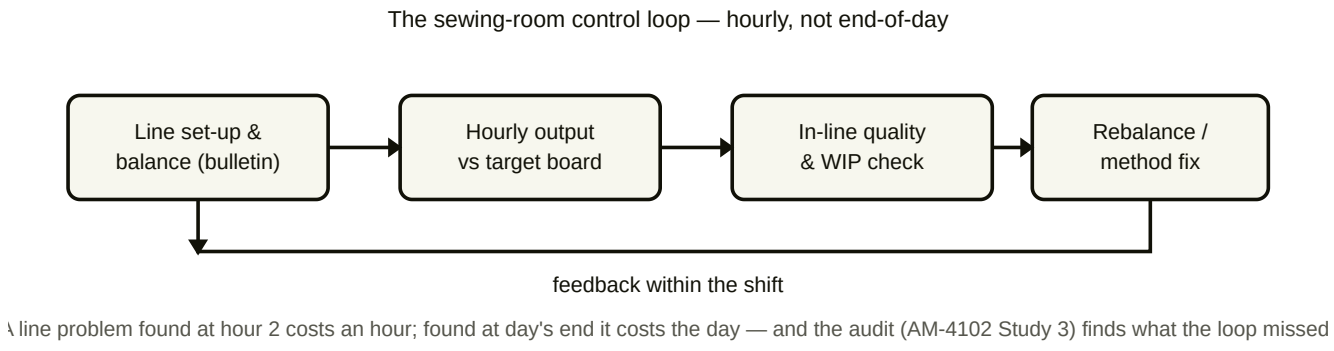


Figure 1 — Sewing-room control is a loop closed every hour: target, measure, diagnose, rebalance.

The sewing room holds most of the workforce and most of the conversion cost, and its management is the daily practice of Topics 1–4: **set-up** from the operation bulletin with machines, attachments and guides ready before operators sit down; **line feeding** so bundles (or hangers) arrive ahead of need in the right sequence — a starved station is lost capacity that no later hour repays; **hourly monitoring** of output against target at the bottleneck and the line end, with the supervisor's interventions recorded; **WIP discipline** — WIP caps per station (Topic 2's Little's-law arithmetic made physical: capped WIP caps throughput time and defect-suspect populations); **operator management** — skill matrix deployment, relief operators for absence, training against the operations each operator can cover; and **housekeeping of quality** — in-line checking at the operation, so the end-of-line gate audits a healthy line instead of discovering a sick one.

4. Where the two systems meet

Materials and sewing meet at three documents: the **trim card** (every accessory the garment needs, with consumption — thread above), the **bundle ticket** (what the line received) and the **daily reconciliation** (cut vs sewn vs finished vs shipped — the cut-to-ship thread of Topic 1). When a garment count disagrees between departments, these records, not memories, settle where it went.

Glossary

TERM	MEANING
Consumption standard	Budgeted material use per garment (fabric m, thread m, buttons pcs) including wastage allowance.
Thread ratio	Metres of thread consumed per metre of seam for a stitch type (≈ 2.5 for 301 at standard density).
Trim card	The order's accessory specification and consumption list: labels, buttons, zips, thread, polybag, carton.
FIFO	First-in-first-out store issue, preventing aged stock and shade-lot mixing.
Skill matrix	Record of which operations each operator can perform and at what efficiency — the deployment tool for balancing and absence cover.

Key points and summary

- Material management: requirement → receive/inspect → store → issue → reconcile, per order; shortage and surplus are equally failures.
- Every material has a consumption standard — thread: seam length × ratio × (1 + wastage) = 17.05 m/garment in the worked example.
- Sewing-room management is an hourly control loop built on the operation bulletin, WIP caps and the skill matrix.
- Trim card, bundle ticket and daily reconciliation are where the two systems audit each other.

Practice

Answer in your own words; brief model answers follow.

1. A garment's total seam length is 7.4 m of 301 stitching (ratio 2.5). With 10% wastage, how many metres of thread per garment, and how many 5,000 m cones for 20,000 garments?

2. An order of 8,000 garments has budgeted fabric consumption 1.45 m/garment. The store issued 11,900 m. Quantify the variance and state the manager's next step.
3. Why does capping WIP per station shorten defect-suspect populations (connect Topic 2)?
4. A line's hourly board shows output on target but end-of-line DHU climbing. What does the sewing-room loop do before the final audit does it for them?
5. Why is fabric issued against the cut plan rather than on the cutting manager's verbal request?

Answer key

1. Thread = 7.4 × 2.5 = 18.5 m; × 1.10 = 20.35 m/garment. Total = 20,000 × 20.35 = 407,000 m → 407,000 ÷ 5,000 = 81.4 → 82 cones.
2. Budget = 8,000 × 1.45 = 11,600 m. Variance = 11,900 − 11,600 = +300 m (2.6% over) — outside a tight allowance: reconcile lays cut, marker used, splice/defect records and remnants returned, per order, and identify whether the loss is marker, spreading or damage before the order closes.
3. Suspect population ≈ production during the throughput window between fault creation and detection. Capped WIP caps throughput time (Little's law), so fewer garments exist between the faulty operation and its discovery.
4. Intervene at the operation: in-line/roving checks identify the defect source now, the supervisor rebalances or fixes method/training, and the affected WIP is sorted while it is still a few bundles — not discovered as a lot failure at audit.
5. Because consumption control is documentary: the cut plan states what the order's markers require; issue against it makes over-issue visible the day it happens. Verbal issue converts the store into an uncontrolled buffer where losses and dead stock accumulate undetected.
